

University of Botswana
Department of Computer Science
Tutorial - Sets

Question 1

Prove that if A and B are sets, then $(A \cup \emptyset) = A$ by showing that:

- $(A \cup \emptyset) \subseteq A$

Question 2

Prove that if A and B are sets, then $(A \cap A) = A$ by showing that:

- $(A \cap A) \subseteq A$

Question 3

Prove that if A and B are sets, then $(\emptyset - A) = \emptyset$ by showing that:

- $(\emptyset - A) \subseteq \emptyset$

Question 4

Prove that if A and B are sets, then $A \cup (A \cap B) = A$ by showing that:

- $A \cup (A \cap B) \subseteq A$

Question 5

Prove that if A and B are sets, then $(A \cup (B - A)) = (A \cup B)$ by showing that:

- $(A \cup (B - A)) \subseteq (A \cup B)$
- $(A \cup B) \subseteq (A \cup (B - A))$

~~**Question 6**~~

~~Prove by mathematical induction that $n^2 + n$ is divisible by 2, $\forall n \geq 1$. Use the template used in class.~~

~~**Question 7**~~

~~Prove by mathematical induction that~~

$$2^n < n! \quad \forall n \geq 4$$

~~Use the template used in class.~~